

Modular Arithmetic and Groups

1 Modulo Means Remainder

Basic Idea

Modulo arithmetic keeps only the remainder.

$$17 \bmod 5 = 2$$

because

$$17 = 5 \cdot 3 + 2.$$

When we work modulo 5, every integer behaves like one of

$$0, 1, 2, 3, 4.$$

Same Remainder

$$17 \equiv 2 \pmod{5}$$

means 17 and 2 have the same remainder after division by 5.

2 Working in $\mathbb{Z}_n$

Meaning of $\mathbb{Z}_n$

$$\mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}$$

It is the set of possible remainders modulo n .

Example:

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}.$$

How $\mathbb{Z}_6$ is calculated

$\mathbb{Z}_6$ means: list all possible remainders after division by 6.

$$\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}.$$

We stop at 5, because the remainder after division by 6 can never be 6.

Examples:

$$8 \bmod 6 = 2, \quad 13 \bmod 6 = 1, \quad 18 \bmod 6 = 0.$$

So 8 behaves like 2, 13 behaves like 1, and 18 behaves like 0 in $\mathbb{Z}_6$.

How to Add and Multiply

Do the normal operation first, then take the remainder.

$$4 + 5 \equiv 3 \pmod{6}$$

because

$$4 + 5 = 9, \quad 9 \bmod 6 = 3.$$

$$4 \cdot 5 \equiv 2 \pmod{6}$$

because

$$4 \cdot 5 = 20, \quad 20 \bmod 6 = 2.$$

3 Group: The Simple Meaning

Group

A **group** is a set where an operation behaves nicely:

the answer stays inside the set,
there is an identity element,
every element has an inverse,
grouping does not change the result.

Word	Simple meaning
Identity	Does nothing. For addition, identity is 0.
Inverse	Cancels an element. For addition, inverse of a is $-a$.
Abelian	Order does not matter. Example: $a + b = b + a$.
Finite	Has a limited number of elements.

4 Addition Modulo n

Always Works

$(\mathbb{Z}_n, +)$ is always a finite abelian group.

Why?

identity = 0.

The inverse of a is the number that makes the sum 0 modulo n .

Example in $\mathbb{Z}_6$

$$4 + 2 \equiv 0 \pmod{6}.$$

So the additive inverse of 4 modulo 6 is 2.

$$-4 \equiv 2 \pmod{6}.$$

5 Multiplication Modulo n

Multiplication is different from addition. Not every number has a multiplicative inverse.

Numbers With Inverses

$$\mathbb{Z}_n^* = \{a : \gcd(a, n) = 1\}$$

These are the numbers that have multiplicative inverses modulo n .

$\mathbb{Z}_n$ vs. $\mathbb{Z}_n^*$

$$\mathbb{Z}_n = \text{all remainders } 0, 1, \dots, n-1.$$

$$\mathbb{Z}_n^* = \text{only the remainders relatively prime to } n.$$

How $\mathbb{Z}_7^*$ is calculated

Start with the nonzero remainders modulo 7:

$$1, 2, 3, 4, 5, 6.$$

Keep the numbers whose gcd with 7 is 1:

$$\gcd(1, 7) = 1, \quad \gcd(2, 7) = 1, \quad \gcd(3, 7) = 1,$$

$$\gcd(4, 7) = 1, \quad \gcd(5, 7) = 1, \quad \gcd(6, 7) = 1.$$

Therefore,

$$\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}.$$

Since 7 is prime, every nonzero remainder modulo 7 is included.

Example: $\mathbb{Z}_{15}^*$

Check the numbers from 1 to 14. Keep only the numbers relatively prime to 15:

$$\mathbb{Z}_{15}^* = \{1, 2, 4, 7, 8, 11, 13, 14\}.$$

Most Important Rule

$$a^{-1} \pmod{n} \text{ exists only if } \gcd(a, n) = 1.$$

If $\gcd(a, n) \neq 1$, then a has no inverse modulo n .

6 Finding a Modular Inverse

Goal

To find the inverse of a modulo n , find a number x such that

$$ax \equiv 1 \pmod{n}.$$

Then x is called the inverse of a modulo n .

Step	What to do
1	Check $\gcd(a, n) = 1$. If not, inverse does not exist.
2	Use Euclid or Extended Euclid to write $1 = ax + ny$.
3	Take the coefficient x . That is the inverse of a .
4	If x is negative, convert it to a positive residue modulo n .
5	Check by multiplying: $a \cdot x \equiv 1 \pmod{n}$.

Example

Find the inverse of 5 modulo 11.

Step 1: Check gcd

$$\gcd(5, 11) = 1.$$

So the inverse exists.

Step 2: Use Euclid

$$11 = 5(2) + 1.$$

Rewrite this to make 1 the subject:

$$1 = 11 - 5(2).$$

In Bezout form,

$$1 = 5(-2) + 11(1).$$

Step 3: Take the coefficient of 5

The coefficient of 5 is -2 . Therefore

$$5(-2) \equiv 1 \pmod{11}.$$

Step 4: Convert to a positive residue

Modulo 11,

$$-2 \equiv 9 \pmod{11}.$$

So

$$5^{-1} \equiv 9 \pmod{11}.$$

Step 5: Check

$$5 \cdot 9 = 45.$$

Since

$$45 \bmod 11 = 1,$$

So

$$5 \cdot 9 \equiv 1 \pmod{11}.$$

Answer

$$5^{-1} \equiv 9 \pmod{11}.$$

No Inverse Case

If the gcd is not 1, stop. For example, $6^{-1} \pmod{15}$ does not exist because

$$\gcd(6, 15) = 3 \neq 1.$$

7 Division Modulo n

Division Means Inverse

In modular arithmetic,

$$\frac{a}{b} \equiv a \cdot b^{-1} \pmod{n}.$$

So division by b is allowed only when b^{-1} exists.

Example:

$$\frac{4}{5} \pmod{11}$$

means

$$4 \cdot 5^{-1} \pmod{11}.$$

Since $5^{-1} \equiv 9 \pmod{11}$,

$$4 \cdot 9 = 36 \equiv 3 \pmod{11}.$$

Therefore,

$$\frac{4}{5} \equiv 3 \pmod{11}.$$

8 Euler's Phi Function

What $\phi(n)$ Counts

$\phi(n)$ = how many numbers from 1 to $n - 1$ are relatively prime to n .

Equivalently,

$$\phi(n) = |\mathbb{Z}_n^*|.$$

Phi Formula

If the different prime divisors of n are known, then

$$\phi(n) = n \prod_{p|n} \left(1 - \frac{1}{p}\right).$$

Example: $\phi(45)$

$$45 = 3^2 \cdot 5.$$

So the different prime divisors are 3 and 5.

$$\begin{aligned}\phi(45) &= 45 \left(1 - \frac{1}{3}\right) \left(1 - \frac{1}{5}\right) \\ &= 45 \cdot \frac{2}{3} \cdot \frac{4}{5} = 24.\end{aligned}$$

$$\phi(45) = 24.$$

Prime Case

If p is prime, then

$$\phi(p) = p - 1.$$

9 Subgroups

Simple Meaning

A **subgroup** is a smaller group inside a bigger group.

Example in $\mathbb{Z}_8$

The full group is

$$\mathbb{Z}_8 = \{0, 1, 2, 3, 4, 5, 6, 7\}.$$

Now take the smaller set

$$S = \{0, 2, 4, 6\}.$$

This is a subgroup because adding two numbers from S , modulo 8, always gives another number from S .

Check

$$2 + 4 = 6,$$

$$4 + 6 = 10 \equiv 2 \pmod{8},$$

$$6 + 6 = 12 \equiv 4 \pmod{8}.$$

All answers are still in

$$\{0, 2, 4, 6\}.$$

Conclusion

$\{0, 2, 4, 6\}$ is a subgroup of $\mathbb{Z}_8$.

Lagrange's Theorem

If a subgroup has size m , and the full group has size N , then

$$m \mid N.$$

Here,

$$|\mathbb{Z}_8| = 8 \quad \text{and} \quad |\{0, 2, 4, 6\}| = 4.$$

Since

$$4 \mid 8,$$

the example matches Lagrange's theorem.

10 Generated Subgroups

Simple Meaning

A **generated subgroup** starts from one element and keeps applying the group operation again and again.

generated subgroup = all elements reached from one starting element.

Example in $\mathbb{Z}_6$

Use addition modulo 6. Start with 2, then keep adding 2:

$$2,$$

$$2 + 2 = 4,$$

$$4 + 2 = 6 \equiv 0 \pmod{6},$$

$$0 + 2 = 2.$$

Now the cycle repeats:

$$2, 4, 0, 2, 4, 0, \dots$$

So

$$\langle 2 \rangle = \{0, 2, 4\}.$$

This means 2 generates $\{0, 2, 4\}$ in $\mathbb{Z}_6$.

Example in $\mathbb{Z}_7^*$

First,

$$\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}.$$

Now start with 3 and keep multiplying by 3, always reducing modulo 7:

$$3^1 \equiv 3, \quad 3^2 \equiv 2, \quad 3^3 \equiv 6,$$

$$3^4 \equiv 4, \quad 3^5 \equiv 5, \quad 3^6 \equiv 1.$$

So

$$\langle 3 \rangle = \{1, 2, 3, 4, 5, 6\}.$$

This means 3 generates all of $\mathbb{Z}_7^*$.

Difference

subgroup = a smaller group inside a bigger group.

generated subgroup = a subgroup created by repeatedly using one element.

Every generated subgroup is a subgroup.

11 Order of an Element

Order

The **order** of an element is how many steps it takes to return to the identity.

In $\mathbb{Z}_7^*$, the identity is 1. From the previous example,

$$3^6 \equiv 1 \pmod{7}.$$

So

$$\text{ord}(3) = 6 \text{ in } \mathbb{Z}_7^*.$$

Final Exam Box

$$\mathbb{Z}_n = \{0, 1, \dots, n-1\}.$$

$(\mathbb{Z}_n, +)$ always works as a group.

$$\mathbb{Z}_n^* = \{a : \gcd(a, n) = 1\}.$$

$a^{-1} \pmod{n}$ exists iff $\gcd(a, n) = 1$.

$$\phi(n) = |\mathbb{Z}_n^*|.$$

subgroup size divides group size.

order = steps needed to return to identity.